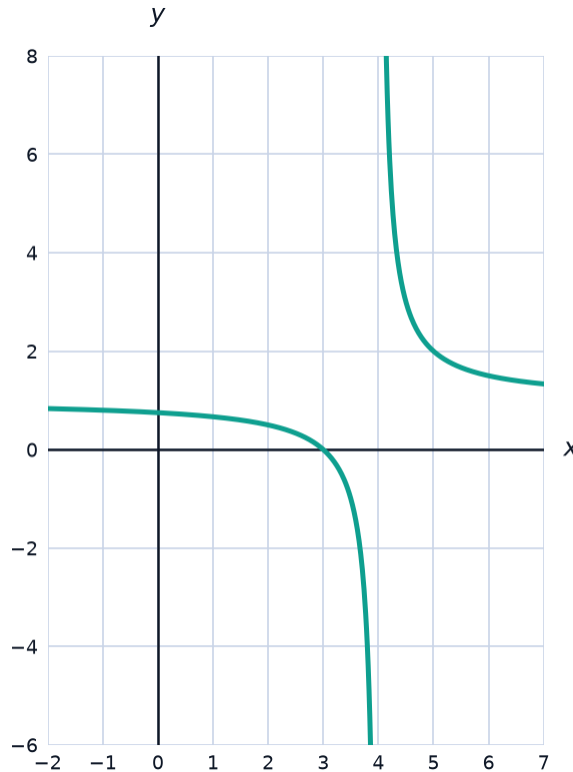


# Rational Functions & Asymptotes



## Vertical asymptotes and holes

- 1 Find the vertical asymptotes and holes of  $f(x) = \frac{x^2 - 9}{x^2 - x - 12}$ .



- 2 Find the y-coordinate of the hole in the function from the previous problem.

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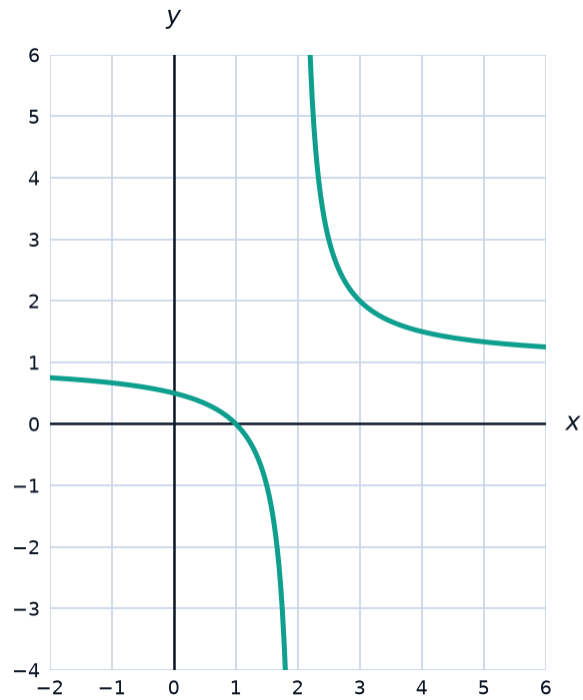
## Horizontal and slant asymptotes

- 3 Find the horizontal asymptote of  $f(x) = \frac{3x^2 - 2x + 1}{5x^2 + 4}$ , using the degrees of numerator and denominator.
- 4 Find the horizontal asymptote of  $g(x) = \frac{2x + 7}{x^2 - 1}$ .
- 5 Explain why  $h(x) = \frac{x^2 + 1}{x - 2}$  has a slant (oblique) asymptote instead of a horizontal one, and find it using polynomial division.

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## Visualizing a rational function's asymptotes

- 6 The graph shows  $f(x) = \frac{1}{x-2} + 1$ , a transformed reciprocal function.



- a Identify the vertical asymptote shown.
- b Identify the horizontal asymptote shown.

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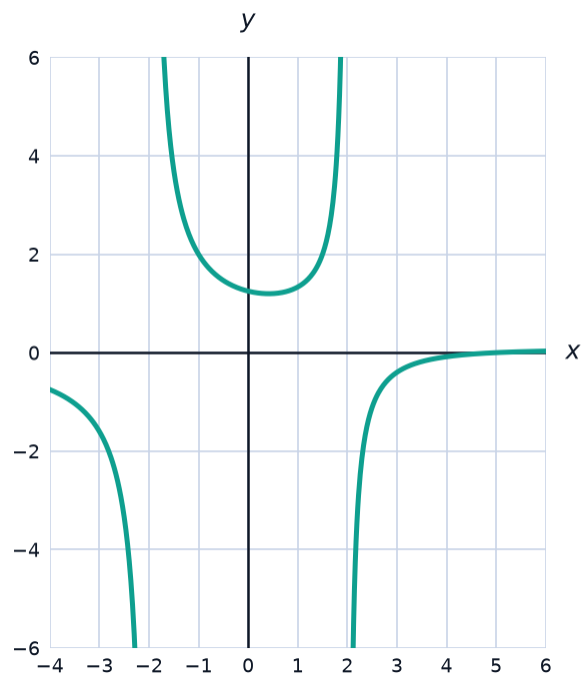
## End behavior of rational functions

- 7 Describe the end behavior of  $f(x) = \frac{2x^3 - x}{x^2 + 1}$  as  $x \rightarrow \pm \infty$  (note the numerator's degree exceeds the denominator's by 1, so the function behaves like its slant asymptote).



### Domain and intercepts

- 8 Find the domain, x-intercept(s), and y-intercept of  $f(x) = \frac{x - 5}{x^2 - 4}$ .



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## Solving rational equations and inequalities

- 9 Solve  $\frac{x+1}{x-3} = 2$ , checking that the solution doesn't create a zero denominator.
- 10 Solve the inequality  $\frac{x-1}{x+2} > 0$  by determining the sign of the expression on each interval created by the zero and the excluded value.

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## Visualizing a function with a slant asymptote

- 11 The graph shows  $f(x) = \frac{x^2}{x-1}$ , which has a slant asymptote.



- a Find the vertical asymptote and confirm it matches the graph.
- b Use polynomial division to find the slant asymptote equation.
- c Explain why this function has no horizontal asymptote.

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## Building a rational function from given features

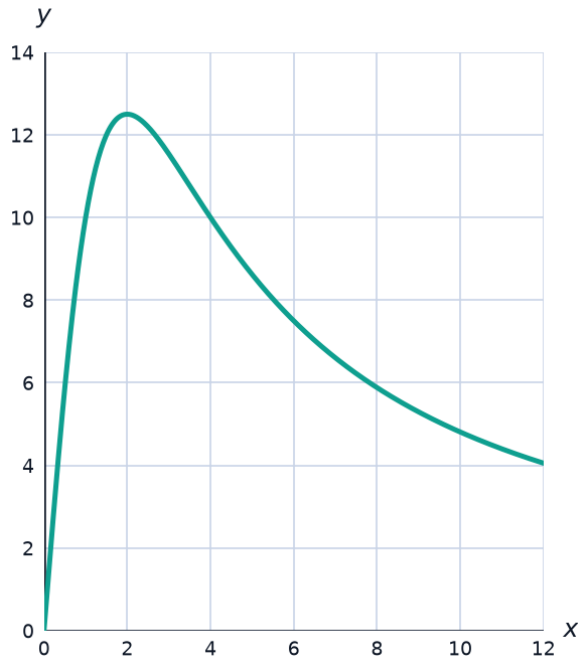
- 12 Construct a rational function with a vertical asymptote at  $x = 3$ , a hole at  $x = -1$ , and a horizontal asymptote at  $y = 2$ .

- 13 Explain why a rational function cannot have both a horizontal asymptote and a slant asymptote at the same time.

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### Rational functions in context

- 14 The concentration of a drug in the bloodstream  $t$  hours after a dose is modeled by  $C(t) = \frac{50t}{t^2 + 4}$  (mg/L). Find the horizontal asymptote and interpret it in context.



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### Multiple choice — easy 15

- 15 Find the horizontal asymptote of  $f(x) = \frac{9x^2 + 5x - 2}{6x^2 + x - 7}$ .

- a.  $y = \frac{2}{3}$
- b.  $y = \frac{5}{2}$
- c.  $y = 0$
- d.  $y = \frac{3}{2}$

16 Find the horizontal asymptote of  $g(x) = \frac{-4x + 6}{5x^2 - 6x - 6}$ .

a.  $y = \frac{4}{5}$

b.  $y = -\frac{4}{5}$

c.  $y = 0$

d.  $y = 1$

17 Find the horizontal asymptote of  $f(x) = \frac{2x^2 - 9x - 9}{2x^2 - 6x - 3}$ .

a.  $y = \frac{3}{2}$

b.  $y = 0$

c.  $y = 1$

d.  $y = 2$

18 Find the horizontal asymptote of  $g(x) = \frac{8x - 4}{7x^2 + 4x + 7}$ .

a.  $y = -\frac{8}{7}$

b.  $y = 1$

c.  $y = 0$

d.  $y = \frac{8}{7}$

19 Find the horizontal asymptote of  $f(x) = \frac{4x^2 - 8x + 6}{9x^2 + 3x + 9}$ .

a.  $y = \frac{9}{4}$

b.  $y = \frac{4}{9}$

c.  $y = 0$

d.  $y = \frac{13}{9}$

20 Find the horizontal asymptote of  $g(x) = \frac{6x - 4}{9x^2 - 5x - 2}$ .

a.  $y = -\frac{2}{3}$

b.  $y = 1$

c.  $y = \frac{2}{3}$

d.  $y = 0$

21 Find the horizontal asymptote of  $f(x) = \frac{6x^2 - 7x}{4x^2 - 7x - 9}$ .

a.  $y = \frac{3}{2}$

b.  $y = \frac{2}{3}$

c.  $y = 0$

d.  $y = \frac{5}{2}$

22 Find the horizontal asymptote of  $f(x) = \frac{5x^2 - 7x - 4}{2x^2 - 5x + 1}$ .

a.  $y = \frac{2}{5}$

b.  $y = \frac{7}{2}$

c.  $y = \frac{5}{2}$

d.  $y = 0$

23 Find the horizontal asymptote of  $f(x) = \frac{4x^2 - 2x + 1}{4x^2 - 4x + 7}$ .

a.  $y = 2$

b.  $y = 1$

c.  $y = 0$

d.  $y = \frac{3}{2}$

24 Find the horizontal asymptote of  $g(x) = \frac{9x + 3}{7x^2 + 7x + 5}$ .

a.  $y = 0$

b.  $y = 1$

c.  $y = \frac{9}{7}$

d.  $y = -\frac{9}{7}$

25 Find the horizontal asymptote of  $g(x) = \frac{x - 5}{3x^2 - 2x + 2}$ .

a.  $y = 0$

b.  $y = -\frac{1}{3}$

c.  $y = \frac{1}{3}$

d.  $y = 1$

26 Find the horizontal asymptote of  $f(x) = \frac{9x^2 + 6x - 3}{8x^2 + 3x - 5}$ .

a.  $y = 0$

b.  $y = \frac{9}{8}$

c.  $y = \frac{8}{9}$

d.  $y = \frac{17}{8}$

27 Find the horizontal asymptote of  $g(x) = \frac{-5x + 4}{3x^2 - x - 1}$ .

a.  $y = 1$

b.  $y = \frac{5}{3}$

c.  $y = -\frac{5}{3}$

d.  $y = 0$



28 Find the horizontal asymptote of  $g(x) = \frac{-7x + 1}{4x^2 - 4x + 4}$ .

a.  $y = \frac{7}{4}$

b.  $y = 0$

c.  $y = -\frac{7}{4}$

d.  $y = 1$

29 Find the horizontal asymptote of  $f(x) = \frac{4x^2 + 6x - 1}{4x^2 + 6x + 2}$ .

a.  $y = 2$

b.  $y = 0$

c.  $y = \frac{1}{2}$

d.  $y = 1$

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### Multiple choice — medium 15

30 Find the hole and vertical asymptote of  $f(x) = \frac{(x + 2)(x - 5)}{(x + 2)(x - 6)}$ .

a. Hole at  $x = 6$ ; vertical asymptote at  $x = -2$ .

b. Hole at  $x = -2$ ; vertical asymptote at  $x = 6$ .

c. Vertical asymptotes at  $x = -2$  and  $x = 6$ , no hole.

d. Hole at  $x = 5$ ; vertical asymptote at  $x = 6$ .

31 Find the hole and vertical asymptote of  $f(x) = \frac{(x + 3)(x - 7)}{(x + 3)(x - 6)}$ .

a. Vertical asymptotes at  $x = -3$  and  $x = 6$ , no hole.

b. Hole at  $x = -3$ ; vertical asymptote at  $x = 6$ .

c. Hole at  $x = 7$ ; vertical asymptote at  $x = 6$ .

d. Hole at  $x = 6$ ; vertical asymptote at  $x = -3$ .

32 Find the hole and vertical asymptote of  $f(x) = \frac{(x+5)(x-6)}{(x+5)(x)}$ .

- a. Vertical asymptotes at  $x = -5$  and  $x = 0$ , no hole.
- b. Hole at  $x = 0$ ; vertical asymptote at  $x = -5$ .
- c. Hole at  $x = -5$ ; vertical asymptote at  $x = 0$ .
- d. Hole at  $x = 6$ ; vertical asymptote at  $x = 0$ .

33 Find the hole and vertical asymptote of  $f(x) = \frac{(x-2)(x-4)}{(x-2)(x+6)}$ .

- a. Hole at  $x = 2$ ; vertical asymptote at  $x = -6$ .
- b. Vertical asymptotes at  $x = 2$  and  $x = -6$ , no hole.
- c. Hole at  $x = -6$ ; vertical asymptote at  $x = 2$ .
- d. Hole at  $x = 4$ ; vertical asymptote at  $x = -6$ .

34 Find the hole and vertical asymptote of  $f(x) = \frac{(x-5)(x+9)}{(x-5)(x+7)}$ .

- a. Hole at  $x = 5$ ; vertical asymptote at  $x = -7$ .
- b. Hole at  $x = -9$ ; vertical asymptote at  $x = -7$ .
- c. Hole at  $x = -7$ ; vertical asymptote at  $x = 5$ .
- d. Vertical asymptotes at  $x = 5$  and  $x = -7$ , no hole.

35 Find the hole and vertical asymptote of  $f(x) = \frac{(x+8)(x-8)}{(x+8)(x-3)}$ .

- a. Vertical asymptotes at  $x = -8$  and  $x = 3$ , no hole.
- b. Hole at  $x = -8$ ; vertical asymptote at  $x = 3$ .
- c. Hole at  $x = 8$ ; vertical asymptote at  $x = 3$ .
- d. Hole at  $x = 3$ ; vertical asymptote at  $x = -8$ .

36 Find the hole and vertical asymptote of  $f(x) = \frac{(x-2)(x-7)}{(x-2)(x+9)}$ .

- a. Hole at  $x = 7$ ; vertical asymptote at  $x = -9$ .
- b. Vertical asymptotes at  $x = 2$  and  $x = -9$ , no hole.
- c. Hole at  $x = -9$ ; vertical asymptote at  $x = 2$ .
- d. Hole at  $x = 2$ ; vertical asymptote at  $x = -9$ .

37 Find the hole and vertical asymptote of  $f(x) = \frac{(x+8)(x-8)}{(x+8)(x-9)}$ .

- a. Hole at  $x = -8$ ; vertical asymptote at  $x = 9$ .
- b. Vertical asymptotes at  $x = -8$  and  $x = 9$ , no hole.
- c. Hole at  $x = 8$ ; vertical asymptote at  $x = 9$ .
- d. Hole at  $x = 9$ ; vertical asymptote at  $x = -8$ .

38 Find the hole and vertical asymptote of  $f(x) = \frac{(x-2)(x+4)}{(x-2)(x-6)}$ .

- a. Hole at  $x = 2$ ; vertical asymptote at  $x = 6$ .
- b. Hole at  $x = 6$ ; vertical asymptote at  $x = 2$ .
- c. Vertical asymptotes at  $x = 2$  and  $x = 6$ , no hole.
- d. Hole at  $x = -4$ ; vertical asymptote at  $x = 6$ .

39 Find the hole and vertical asymptote of  $f(x) = \frac{(x-3)(x-6)}{(x-3)(x)}$ .

- a. Hole at  $x = 6$ ; vertical asymptote at  $x = 0$ .
- b. Vertical asymptotes at  $x = 3$  and  $x = 0$ , no hole.
- c. Hole at  $x = 3$ ; vertical asymptote at  $x = 0$ .
- d. Hole at  $x = 0$ ; vertical asymptote at  $x = 3$ .

**40** Find the hole and vertical asymptote of  $f(x) = \frac{(x-1)(x+1)}{(x-1)(x)}$ .

- a.** Hole at  $x = 1$ ; vertical asymptote at  $x = 0$ .
- b.** Vertical asymptotes at  $x = 1$  and  $x = 0$ , no hole.
- c.** Hole at  $x = 0$ ; vertical asymptote at  $x = 1$ .
- d.** Hole at  $x = -1$ ; vertical asymptote at  $x = 0$ .

**41** Find the hole and vertical asymptote of  $f(x) = \frac{(x+5)(x+8)}{(x+5)(x+2)}$ .

- a.** Hole at  $x = -8$ ; vertical asymptote at  $x = -2$ .
- b.** Vertical asymptotes at  $x = -5$  and  $x = -2$ , no hole.
- c.** Hole at  $x = -2$ ; vertical asymptote at  $x = -5$ .
- d.** Hole at  $x = -5$ ; vertical asymptote at  $x = -2$ .

**42** Find the hole and vertical asymptote of  $f(x) = \frac{(x-6)(x+1)}{(x-6)(x-5)}$ .

- a.** Hole at  $x = -1$ ; vertical asymptote at  $x = 5$ .
- b.** Vertical asymptotes at  $x = 6$  and  $x = 5$ , no hole.
- c.** Hole at  $x = 6$ ; vertical asymptote at  $x = 5$ .
- d.** Hole at  $x = 5$ ; vertical asymptote at  $x = 6$ .

**43** Find the hole and vertical asymptote of  $f(x) = \frac{(x+8)(x)}{(x+8)(x+3)}$ .

- a.** Hole at  $x = -3$ ; vertical asymptote at  $x = -8$ .
- b.** Hole at  $x = -8$ ; vertical asymptote at  $x = -3$ .
- c.** Hole at  $x = 0$ ; vertical asymptote at  $x = -3$ .
- d.** Vertical asymptotes at  $x = -8$  and  $x = -3$ , no hole.

- 44 Find the hole and vertical asymptote of  $f(x) = \frac{(x+6)(x+4)}{(x+6)(x+7)}$ .
- a. Vertical asymptotes at  $x = -6$  and  $x = -7$ , no hole.
  - b. Hole at  $x = -4$ ; vertical asymptote at  $x = -7$ .
  - c. Hole at  $x = -7$ ; vertical asymptote at  $x = -6$ .
  - d. Hole at  $x = -6$ ; vertical asymptote at  $x = -7$ .
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### Multiple choice — hard 15

- 45 Find the slant asymptote of  $h(x) = \frac{x^2 + x + 7}{x - 7}$ .
- a.  $y = x + 9$
  - b.  $y = x + 7$
  - c.  $y = x - 8$
  - d.  $y = x + 8$
- 46 Find the slant asymptote of  $h(x) = \frac{x^2 + 8x - 7}{x + 9}$ .
- a.  $y = x$
  - b.  $y = x - 2$
  - c.  $y = x + 1$
  - d.  $y = x - 1$
- 47 Find the slant asymptote of  $h(x) = \frac{x^2 + 8x - 5}{x + 1}$ .
- a.  $y = x - 7$
  - b.  $y = x + 8$
  - c.  $y = x + 6$
  - d.  $y = x + 7$

48 Find the slant asymptote of  $h(x) = \frac{x^2 - 8x - 3}{x - 8}$ .

a.  $y = x$

b.  $y = x + 2$

c.  $y = x + 1$

d.  $y = x - 1$

49 Find the slant asymptote of  $h(x) = \frac{x^2 + 4x + 5}{x + 6}$ .

a.  $y = x - 3$

b.  $y = x - 1$

c.  $y = x - 2$

d.  $y = x + 2$

50 Solve  $\frac{x - 9}{x + 7} = 2$  for  $x$ .

a.  $-23$

b.  $23$

c.  $-\frac{9}{2}$

d.  $-22$

51 Solve  $\frac{x + 4}{x - 3} = -3$  for  $x$ .

a.  $-\frac{5}{4}$

b.  $\frac{5}{4}$

c.  $-\frac{4}{3}$

d.  $\frac{9}{4}$

52 Solve  $\frac{x-8}{x-1} = 5$  for  $x$ .

a.  $-\frac{8}{5}$

b.  $-\frac{3}{4}$

c.  $\frac{3}{4}$

d.  $\frac{1}{4}$

53 Solve  $\frac{x+3}{x} = -2$  for  $x$ .

a. 1

b.  $-\frac{3}{2}$

c. 0

d. -1

54 Solve  $\frac{x-5}{x+7} = 4$  for  $x$ .

a. 11

b.  $-\frac{5}{4}$

c. -11

d. -10

55 Solve the inequality  $\frac{x+6}{x-3} > 0$ .

a.  $(-\infty, 3) \cup (-6, \infty)$

b.  $(-\infty, -6) \cup (3, \infty)$

c.  $[-6, 3]$

d.  $(-6, 3)$

56 Solve the inequality  $\frac{x-6}{x-7} > 0$ .

a.  $(-\infty, 6) \cup (7, \infty)$

b.  $[6, 7]$

c.  $(6, 7)$

d.  $(-\infty, 7) \cup (6, \infty)$

57 Solve the inequality  $\frac{x-7}{x-5} > 0$ .

a.  $(-\infty, 7) \cup (5, \infty)$

b.  $[5, 7]$

c.  $(5, 7)$

d.  $(-\infty, 5) \cup (7, \infty)$

58 Solve the inequality  $\frac{x-2}{x-7} > 0$ .

a.  $(-\infty, 7) \cup (2, \infty)$

b.  $(2, 7)$

c.  $(-\infty, 2) \cup (7, \infty)$

d.  $[2, 7]$

59 Solve the inequality  $\frac{x+8}{x-7} > 0$ .

a.  $[-8, 7]$

b.  $(-\infty, -8) \cup (7, \infty)$

c.  $(-\infty, 7) \cup (-8, \infty)$

d.  $(-8, 7)$